Nonconcave moduli in the Whitney-Glaeser theorem

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Source and target

The source paper is Johanis, Krystof, and Zajicek, *On Whitney-type extension theorems on Banach spaces for $C^{1,\omega}$, $C^{1,+}$, $C^{1,+}_{\mathrm{loc}}$, and $C^{1,+}_{\mathrm{B}}$-smooth functions*, arXiv:2305.19995 [1]. Its Theorem `t:ext_C1om` proves a Whitney-Glaeser extension theorem for concave finite moduli ω. In Remark `r:moduli_nonconc`, the authors explain that the theorem also holds for some nonconcave moduli: it is enough that there are a concave modulus ψ and a constant $K > 0$ such that

$$\psi \le \omega \le K\psi.$$

They then ask whether this condition is necessary.

Result

The answer is yes, for the sufficiency direction with the derivative interpolation conclusion of Theorem `t:ext_C1om`. Thus the sufficient condition in Remark `r:moduli_nonconc`(b) is also necessary.

Theorem 1. *Let $\omega \in \mathcal{M}$ be a nonzero finite modulus. Consider the following one-dimensional Whitney-Glaeser principle $P_\omega(\mathbb{R})$: for every $E \subset \mathbb{R}$, every $f : E \to \mathbb{R}$, and every $G : E \to \mathbb{R}$ satisfying condition $\mathcal{WG}\omega$, there is $F \in C^{1,\omega}(\mathbb{R})$ such that $F = f$ and $F' = G$ on E.*

Then $P_\omega(\mathbb{R})$ holds if and only if there are a concave modulus $\psi \in \mathcal{M}$ and a constant $K > 0$ such that

$$\psi \le \omega \le K\psi.$$

Consequently, the source theorem can hold with derivative interpolation on its Banach-space class only under this same condition, since that class contains the line.

Proof Intuition

To force a lower bound on any possible extension modulus, put a very sparse Whitney datum on a line. At dyadic scale 2^k, use two points distance 2^k apart, set both function values equal to zero, and prescribe derivative values whose jump is exactly $\omega(2^k)$. The pairs are placed so far apart that all cross-pair Whitney inequalities are harmless.

If an extension $F \in C^{1,\omega}$ interpolates these derivatives, the minimal modulus m of F' satisfies

$$m(2^k) \ge \omega(2^k) \quad (k \in \mathbb{Z}), \qquad m \le C\omega.$$

The modulus m is subadditive. Stechkin's concavification lemma then produces a concave θ with $m \le \theta \le 2m$. The dyadic lower bound and concavity imply $\theta(t) \ge \frac{1}{2}\omega(t)$ for every $t > 0$, while $\theta \le 2C\omega$. A rescaling of θ is the desired concave modulus comparable to ω.

Proof

The sufficiency direction is exactly the observation made in the source paper: if f satisfies $\mathcal{WG}\omega$ and $\psi \le \omega \le K\psi$ with ψ concave, then f satisfies $\mathcal{WG}\psi$, after increasing the Whitney constant. The concave case of Theorem `t:ext_C1om` gives an extension $F \in C^{1,\psi}(X) \subset C^{1,\omega}(X)$, with $DF = G$ on E.

It remains to prove necessity. Suppose $P_\omega(\mathbb{R})$ holds. The source paper already observes in Remark `r:moduli_nonconc`(a) that this implies that ω dominates a nonzero concave modulus. In particular $\omega(t) > 0$ for every $t > 0$.

Set $r_k = 2^k$, $k \in \mathbb{Z}$. We first choose pair locations

$$a_k < b_k, \qquad b_k - a_k = r_k,$$

with the following separation property: there is $M \ge 1$ such that whenever $k \ne j$, $u \in \{a_k, b_k\}$, and $v \in \{a_j, b_j\}$,

$$\max\{\omega(r_k), \omega(r_j)\} \le M\,\omega(|u - v|). \tag{1}$$

This is immediate by induction. If ω is unbounded, take $M = 1$ and place each new pair far enough from the finitely many old pairs. If ω is bounded, let $L = \sup_{t \ge 0} \omega(t)$, choose $d_0 > 0$, and take $M \ge L/\omega(d_0)$; placing distinct pairs at distance at least d_0 gives (1).

Let $X = \mathbb{R}$, let

$$E = \{a_k, b_k : k \in \mathbb{Z}\},$$

and define

$$f \equiv 0, \qquad G(a_k) = 0, \qquad G(b_k) = \omega(r_k)$$

where $G(x) \in X^* \cong \mathbb{R}$. We claim that this datum satisfies $\mathcal{WG}\omega$. If x, y are the two points of the same dyadic pair, then $|x - y| = r_k$ and the required estimates hold with constant $\max\{M, 1\}$. If x, y belong to different pairs, (1) gives

$$|G(x) - G(y)| \le M\omega(|x - y|)$$

and also

$$|f(y) - f(x) - G(x)(y - x)| = |G(x)|\,|y - x| \le M\omega(|x - y|)\,|x - y|.$$

Thus f satisfies $\mathcal{WG}\omega$ on E.

By $P_\omega(\mathbb{R})$, there exists $F \in C^{1,\omega}(\mathbb{R})$ such that

$$F = f, \qquad F' = G \quad \text{on } E.$$

Let m be the minimal modulus of continuity of F':

$$m(t) = \sup\{|F'(s) - F'(u)| : |s - u| \le t\}.$$

Since F' is uniformly continuous on the convex set $\mathbb{R}$, m is a finite subadditive modulus. Also, because $F \in C^{1,\omega}(\mathbb{R})$, there is $C > 0$ such that

$$m(t) \le C\omega(t) \qquad (t \ge 0).$$

On the other hand, for each $k \in \mathbb{Z}$,

$$m(r_k) \ge |F'(b_k) - F'(a_k)| = \omega(r_k).$$

By Stechkin's concavification lemma, as quoted in Lemma `l:Steckin`(iii) of the source paper, there is a concave modulus θ such that

$$m \leq \theta \leq 2m.$$

Hence $\theta \leq 2C\omega$. For the reverse estimate, fix $t > 0$ and choose $k \in \mathbb{Z}$ with $r_k \leq t < r_{k+1} = 2r_k$. Since θ is concave, nondecreasing, and $\theta(0) = 0$,

$$\theta(t) \geq \frac{t}{r_{k+1}}\theta(r_{k+1}) \geq \frac{1}{2}\omega(r_{k+1}) \geq \frac{1}{2}\omega(t).$$

Therefore

$$\frac{1}{2}\omega \leq \theta \leq 2C\omega.$$

Putting $\psi = (2C)^{-1}\theta$, we get a concave modulus with

$$\psi \leq \omega \leq 4C\,\psi.$$

This proves the necessity and hence the theorem.

Scope and verification notes

This packet answers only Remark `r:moduli_nonconc`; it does not address the paper's other open remarks on relaxing the smoothness assumption on X, removing the extra hypotheses in Corollary `c:extC1+-quasic`, or extensions from bounded convex sets in nonsuperreflexive spaces.

The proof uses only one-dimensional data, the derivative-interpolation part of the Whitney-Glaeser conclusion, and the same Stechkin concavification lemma already used in the source paper. Local index search found no existing packet or attempt for arXiv:2305.19995. Local parsed-source search found only the original open remark. Web search on 2026-06-18 found the source paper and the later arXiv:2507.09384 follow-up, but no answer to this nonconcave-modulus question.

References

[1] M. Johanis, V. Krystof, and L. Zajicek. *On Whitney-type extension theorems on Banach spaces for $C^{1,\omega}$, $C^{1,+}$, $C^{1,+}_{\mathrm{loc}}$, and $C^{1,+}_{\mathrm{B}}$-smooth functions.* arXiv:2305.19995, 2023.